GloVe: Global Vectors for Word Representation

Jeffrey Pennington, Richard Socher, Christopher D. Manning
Computer Science Department, Stanford University, Stanford, CA 94305
jpennin@stanford.edu, richard@socher.org, manning@stanford.edu

Abstract

Recent methods for learning vector space representations of words have succeeded in capturing fine-grained semantic and syntactic regularities using vector arithmetic, but the origin of these regularities has remained opaque. We analyze and make explicit the model properties needed for such regularities to emerge in word vectors. The result is a new global log-bilinear regression model that combines the advantages of the two major model families in the literature: global matrix factorization and local context window methods. Our model efficiently leverages statistical information by training only on the nonzero elements in a word-word co-occurrence matrix, rather than on the entire sparse matrix or on individual context windows in a large corpus. The model produces a vector space with meaningful substructure, as evidenced by its performance of 75% on a recent word analogy task. It also outperforms related models on similarity tasks and named entity recognition.

1 Introduction

Semantic vector space models of language represent each word with a real-valued vector. These vectors can be used as features in a variety of applications, such as information retrieval (Manning et al., 2008), document classification (Sebastiani, 2002), question answering (Tellex et al., 2003), named entity recognition (Turian et al., 2010), and parsing (Socher et al., 2013).

Most word vector methods rely on the distance or angle between pairs of word vectors as the primary method for evaluating the intrinsic quality of such a set of word representations. Recently, Mikolov et al. (2013c) introduced a new evaluation scheme based on word analogies that probes the finer structure of the word vector space by examining not the scalar distance between word vectors, but rather their various dimensions of difference. For example, the analogy "king is to queen as man is to woman" should be encoded in the vector space by the vector equation $king - queen = man - woman$. This evaluation scheme favors models that produce dimensions of meaning, thereby capturing the multi-clustering idea of distributed representations (Bengio, 2009).

The two main model families for learning word vectors are: 1) global matrix factorization methods, such as latent semantic analysis (LSA) (Deerwester et al., 1990) and 2) local context window methods, such as the skip-gram model of Mikolov et al. (2013c). Currently, both families suffer significant drawbacks. While methods like LSA efficiently leverage statistical information, they do relatively poorly on the word analogy task, indicating a sub-optimal vector space structure. Methods like skip-gram may do better on the analogy task, but they poorly utilize the statistics of the corpus since they train on separate local context windows instead of on global co-occurrence counts.

In this work, we analyze the model properties necessary to produce linear directions of meaning and argue that global log-bilinear regression models are appropriate for doing so. We propose a specific weighted least squares model that trains on global word-word co-occurrence counts and thus makes efficient use of statistics. The model produces a word vector space with meaningful substructure, as evidenced by its state-of-the-art performance of 75% accuracy on the word analogy dataset. We also demonstrate that our methods outperform other current methods on several word similarity tasks, and also on a common named entity recognition (NER) benchmark.

We provide the source code for the model as well as trained word vectors at `http://nlp.stanford.edu/projects/glove/`.

2 Related Work

Matrix Factorization Methods. Matrix factorization methods for generating low-dimensional word representations have roots stretching as far back as LSA. These methods utilize low-rank approximations to decompose large matrices that capture statistical information about a corpus. The particular type of information captured by such matrices varies by application. In LSA, the matrices are of "term-document" type, i.e., the rows correspond to words or terms, and the columns correspond to different documents in the corpus. In contrast, the Hyperspace Analogue to Language (HAL) (Lund and Burgess, 1996), for example, utilizes matrices of "term-term" type, i.e., the rows and columns correspond to words and the entries correspond to the number of times a given word occurs in the context of another given word.

A main problem with HAL and related methods is that the most frequent words contribute a disproportionate amount to the similarity measure: the number of times two words co-occur with *the* or *and*, for example, will have a large effect on their similarity despite conveying relatively little about their semantic relatedness. A number of techniques exist that addresses this shortcoming of HAL, such as the COALS method (Rohde et al., 2006), in which the co-occurrence matrix is first transformed by an entropy- or correlation-based normalization. An advantage of this type of transformation is that the raw co-occurrence counts, which for a reasonably sized corpus might span 8 or 9 orders of magnitude, are compressed so as to be distributed more evenly in a smaller interval. A variety of newer models also pursue this approach, including a study (Bullinaria and Levy, 2007) that indicates that positive pointwise mutual information (PPMI) is a good transformation. More recently, a square root type transformation in the form of Hellinger PCA (HPCA) (Lebret and Collobert, 2014) has been suggested as an effective way of learning word representations.

Shallow Window-Based Methods. Another approach is to learn word representations that aid in making predictions within local context windows. For example, Bengio et al. (2003) introduced a model that learns word vector representations as part of a simple neural network architecture for language modeling. Collobert and Weston (2008) decoupled the word vector training from the downstream training objectives, which paved the way for Collobert et al. (2011) to use the full context of a word for learning the word representations, rather than just the preceding context as is the case with language models.

Recently, the importance of the full neural network structure for learning useful word representations has been called into question. The skip-gram and continuous bag-of-words (CBOW) models of Mikolov et al. (2013a) propose a simple single-layer architecture based on the inner product between two word vectors. Mnih and Kavukcuoglu (2013) also proposed closely-related vector log-bilinear models, vLBL and ivLBL, and Levy et al. (2014) proposed explicit word embeddings based on a PPMI metric.

In the skip-gram and ivLBL models, the objective is to predict a word's context given the word itself, whereas the objective in the CBOW and vLBL models is to predict a word given its context. Through evaluation on a word analogy task, these models demonstrated the capacity to learn linguistic patterns as linear relationships between the word vectors.

Unlike the matrix factorization methods, the shallow window-based methods suffer from the disadvantage that they do not operate directly on the co-occurrence statistics of the corpus. Instead, these models scan context windows across the entire corpus, which fails to take advantage of the vast amount of repetition in the data.

3 The GloVe Model

The statistics of word occurrences in a corpus is the primary source of information available to all unsupervised methods for learning word representations, and although many such methods now exist, the question still remains as to how meaning is generated from these statistics, and how the resulting word vectors might represent that meaning. In this section, we shed some light on this question. We use our insights to construct a new model for word representation which we call GloVe, for Global Vectors, because the global corpus statistics are captured directly by the model.

First we establish some notation. Let the matrix of word-word co-occurrence counts be denoted by X, whose entries X_{ij} tabulate the number of times word j occurs in the context of word i. Let $X_i = \sum_k X_{ik}$ be the number of times any word appears in the context of word i. Finally, let $P_{ij} = P(j|i) = X_{ij}/X_i$ be the probability that word j appear in the

Table 1: Co-occurrence probabilities for target words *ice* and *steam* with selected context words from a 6 billion token corpus. Only in the ratio does noise from non-discriminative words like *water* and *fashion* cancel out, so that large values (much greater than 1) correlate well with properties specific to ice, and small values (much less than 1) correlate well with properties specific of steam.

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k\|ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k\|steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k\|ice)/P(k\|steam)$	8.9	8.5×10^{-2}	1.36	0.96

context of word i.

We begin with a simple example that showcases how certain aspects of meaning can be extracted directly from co-occurrence probabilities. Consider two words i and j that exhibit a particular aspect of interest; for concreteness, suppose we are interested in the concept of thermodynamic phase, for which we might take $i = ice$ and $j = steam$. The relationship of these words can be examined by studying the ratio of their co-occurrence probabilities with various probe words, k. For words k related to ice but not steam, say $k = solid$, we expect the ratio P_{ik}/P_{jk} will be large. Similarly, for words k related to steam but not ice, say $k = gas$, the ratio should be small. For words k like *water* or *fashion*, that are either related to both ice and steam, or to neither, the ratio should be close to one. Table 1 shows these probabilities and their ratios for a large corpus, and the numbers confirm these expectations. Compared to the raw probabilities, the ratio is better able to distinguish relevant words (*solid* and *gas*) from irrelevant words (*water* and *fashion*) and it is also better able to discriminate between the two relevant words.

The above argument suggests that the appropriate starting point for word vector learning should be with ratios of co-occurrence probabilities rather than the probabilities themselves. Noting that the ratio P_{ik}/P_{jk} depends on three words i, j, and k, the most general model takes the form,

$$F(w_i, w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}}, \tag{1}$$

where $w \in \mathbb{R}^d$ are word vectors and $\tilde{w} \in \mathbb{R}^d$ are separate context word vectors whose role will be discussed in Section 4.2. In this equation, the right-hand side is extracted from the corpus, and F may depend on some as-of-yet unspecified parameters. The number of possibilities for F is vast, but by enforcing a few desiderata we can select a unique choice. First, we would like F to encode the information present the ratio P_{ik}/P_{jk} in the word vector space. Since vector spaces are inherently linear structures, the most natural way to do this is with vector differences. With this aim, we can restrict our consideration to those functions F that depend only on the difference of the two target words, modifying Eqn. (1) to,

$$F(w_i - w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}}. \tag{2}$$

Next, we note that the arguments of F in Eqn. (2) are vectors while the right-hand side is a scalar. While F could be taken to be a complicated function parameterized by, e.g., a neural network, doing so would obfuscate the linear structure we are trying to capture. To avoid this issue, we can first take the dot product of the arguments,

$$F\left((w_i - w_j)^T \tilde{w}_k\right) = \frac{P_{ik}}{P_{jk}}, \tag{3}$$

which prevents F from mixing the vector dimensions in undesirable ways. Next, note that for word-word co-occurrence matrices, the distinction between a word and a context word is arbitrary and that we are free to exchange the two roles. To do so consistently, we must not only exchange $w \leftrightarrow \tilde{w}$ but also $X \leftrightarrow X^T$. Our final model should be invariant under this relabeling, but Eqn. (3) is not. However, the symmetry can be restored in two steps. First, we require that F be a homomorphism between the groups $(\mathbb{R}, +)$ and $(\mathbb{R}_{>0}, \times)$, i.e.,

$$F\left((w_i - w_j)^T \tilde{w}_k\right) = \frac{F(w_i^T \tilde{w}_k)}{F(w_j^T \tilde{w}_k)}, \tag{4}$$

which, by Eqn. (3), is solved by,

$$F(w_i^T \tilde{w}_k) = P_{ik} = \frac{X_{ik}}{X_i}. \tag{5}$$

The solution to Eqn. (4) is $F = \exp$, or,

$$w_i^T \tilde{w}_k = \log(P_{ik}) = \log(X_{ik}) - \log(X_i). \tag{6}$$

Next, we note that Eqn. (6) would exhibit the exchange symmetry if not for the $\log(X_i)$ on the right-hand side. However, this term is independent of k so it can be absorbed into a bias b_i for w_i. Finally, adding an additional bias $\tilde{b}_k$ for $\tilde{w}_k$ restores the symmetry,

$$w_i^T \tilde{w}_k + b_i + \tilde{b}_k = \log(X_{ik}) \,. \tag{7}$$

Eqn. (7) is a drastic simplification over Eqn. (1), but it is actually ill-defined since the logarithm diverges whenever its argument is zero. One resolution to this issue is to include an additive shift in the logarithm, $\log(X_{ik}) \rightarrow \log(1 + X_{ik})$, which maintains the sparsity of X while avoiding the divergences. The idea of factorizing the log of the co-occurrence matrix is closely related to LSA and we will use the resulting model as a baseline in our experiments. A main drawback to this model is that it weighs all co-occurrences equally, even those that happen rarely or never. Such rare co-occurrences are noisy and carry less information than the more frequent ones — yet even just the zero entries account for 75–95% of the data in X, depending on the vocabulary size and corpus.

We propose a new weighted least squares regression model that addresses these problems. Casting Eqn. (7) as a least squares problem and introducing a weighting function $f(X_{ij})$ into the cost function gives us the model

$$J = \sum_{i,j=1}^{V} f\left(X_{ij}\right)\left(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij}\right)^2 \,, \tag{8}$$

where V is the size of the vocabulary. The weighting function should obey the following properties:

1. $f(0) = 0$. If f is viewed as a continuous function, it should vanish as $x \rightarrow 0$ fast enough that the $\lim_{x\rightarrow 0} f(x)\log^2 x$ is finite.

2. $f(x)$ should be non-decreasing so that rare co-occurrences are not overweighted.

3. $f(x)$ should be relatively small for large values of x, so that frequent co-occurrences are not overweighted.

Of course a large number of functions satisfy these properties, but one class of functions that we found to work well can be parameterized as,

$$f(x) = \begin{cases} (x/x_{\max})^\alpha & \text{if } x < x_{\max} \\ 1 & \text{otherwise} \end{cases} \,. \tag{9}$$

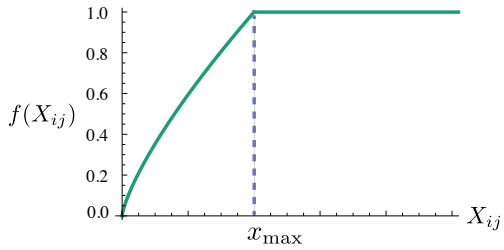


Figure 1: Weighting function f with $\alpha = 3/4$.

The performance of the model depends weakly on the cutoff, which we fix to $x_{\max} = 100$ for all our experiments. We found that $\alpha = 3/4$ gives a modest improvement over a linear version with $\alpha = 1$. Although we offer only empirical motivation for choosing the value 3/4, it is interesting that a similar fractional power scaling was found to give the best performance in (Mikolov et al., 2013a).

3.1 Relationship to Other Models

Because all unsupervised methods for learning word vectors are ultimately based on the occurrence statistics of a corpus, there should be commonalities between the models. Nevertheless, certain models remain somewhat opaque in this regard, particularly the recent window-based methods like skip-gram and ivLBL. Therefore, in this subsection we show how these models are related to our proposed model, as defined in Eqn. (8).

The starting point for the skip-gram or ivLBL methods is a model Q_{ij} for the probability that word j appears in the context of word i. For concreteness, let us assume that Q_{ij} is a softmax,

$$Q_{ij} = \frac{\exp(w_i^T \tilde{w}_j)}{\sum_{k=1}^{V} \exp(w_i^T \tilde{w}_k)} \,. \tag{10}$$

Most of the details of these models are irrelevant for our purposes, aside from the the fact that they attempt to maximize the log probability as a context window scans over the corpus. Training proceeds in an on-line, stochastic fashion, but the implied global objective function can be written as,

$$J = -\sum_{\substack{i \in \text{corpus} \\ j \in \text{context}(i)}} \log Q_{ij} \,. \tag{11}$$

Evaluating the normalization factor of the softmax for each term in this sum is costly. To allow for efficient training, the skip-gram and ivLBL models introduce approximations to Q_{ij}. However, the sum in Eqn. (11) can be evaluated much

more efficiently if we first group together those terms that have the same values for i and j,

$$J = -\sum_{i=1}^{V}\sum_{j=1}^{V} X_{ij} \log Q_{ij} \,, \tag{12}$$

where we have used the fact that the number of like terms is given by the co-occurrence matrix X.

Recalling our notation for $X_i = \sum_k X_{ik}$ and $P_{ij} = X_{ij}/X_i$, we can rewrite J as,

$$J = -\sum_{i=1}^{V} X_i \sum_{j=1}^{V} P_{ij} \log Q_{ij} = \sum_{i=1}^{V} X_i H(P_i, Q_i) \,, \tag{13}$$

where $H(P_i, Q_i)$ is the cross entropy of the distributions P_i and Q_i, which we define in analogy to X_i. As a weighted sum of cross-entropy error, this objective bears some formal resemblance to the weighted least squares objective of Eqn. (8). In fact, it is possible to optimize Eqn. (13) directly as opposed to the on-line training methods used in the skip-gram and ivLBL models. One could interpret this objective as a "global skip-gram" model, and it might be interesting to investigate further. On the other hand, Eqn. (13) exhibits a number of undesirable properties that ought to be addressed before adopting it as a model for learning word vectors.

To begin, cross entropy error is just one among many possible distance measures between probability distributions, and it has the unfortunate property that distributions with long tails are often modeled poorly with too much weight given to the unlikely events. Furthermore, for the measure to be bounded it requires that the model distribution Q be properly normalized. This presents a computational bottleneck owing to the sum over the whole vocabulary in Eqn. (10), and it would be desirable to consider a different distance measure that did not require this property of Q. A natural choice would be a least squares objective in which normalization factors in Q and P are discarded,

$$\hat{J} = \sum_{i,j} X_i \left(\hat{P}_{ij} - \hat{Q}_{ij}\right)^2 \tag{14}$$

where $\hat{P}_{ij} = X_{ij}$ and $\hat{Q}_{ij} = \exp(w_i^T \tilde{w}_j)$ are the unnormalized distributions. At this stage another problem emerges, namely that X_{ij} often takes very large values, which can complicate the optimization. An effective remedy is to minimize the squared error of the logarithms of $\hat{P}$ and $\hat{Q}$ instead,

$$\begin{aligned} \hat{J} &= \sum_{i,j} X_i \left(\log \hat{P}_{ij} - \log \hat{Q}_{ij}\right)^2 \\ &= \sum_{i,j} X_i \left(w_i^T \tilde{w}_j - \log X_{ij}\right)^2 \,. \end{aligned} \tag{15}$$

Finally, we observe that while the weighting factor X_i is preordained by the on-line training method inherent to the skip-gram and ivLBL models, it is by no means guaranteed to be optimal. In fact, Mikolov et al. (2013a) observe that performance can be increased by filtering the data so as to reduce the effective value of the weighting factor for frequent words. With this in mind, we introduce a more general weighting function, which we are free to take to depend on the context word as well. The result is,

$$\hat{J} = \sum_{i,j} f(X_{ij}) \left(w_i^T \tilde{w}_j - \log X_{ij}\right)^2 \,, \tag{16}$$

which is equivalent[1] to the cost function of Eqn. (8), which we derived previously.

3.2 Complexity of the model

As can be seen from Eqn. (8) and the explicit form of the weighting function $f(X)$, the computational complexity of the model depends on the number of nonzero elements in the matrix X. As this number is always less than the total number of entries of the matrix, the model scales no worse than $O(|V|^2)$. At first glance this might seem like a substantial improvement over the shallow window-based approaches, which scale with the corpus size, $|C|$. However, typical vocabularies have hundreds of thousands of words, so that $|V|^2$ can be in the hundreds of billions, which is actually much larger than most corpora. For this reason it is important to determine whether a tighter bound can be placed on the number of nonzero elements of X.

In order to make any concrete statements about the number of nonzero elements in X, it is necessary to make some assumptions about the distribution of word co-occurrences. In particular, we will assume that the number of co-occurrences of word i with word j, X_{ij}, can be modeled as a power-law function of the frequency rank of that word pair, r_{ij}:

$$X_{ij} = \frac{k}{(r_{ij})^{\alpha}} \,. \tag{17}$$

[1] We could also include bias terms in Eqn. (16).

The total number of words in the corpus is proportional to the sum over all elements of the co-occurrence matrix X,

$$|C| \sim \sum_{ij} X_{ij} = \sum_{r=1}^{|X|} \frac{k}{r^\alpha} = k H_{|X|,\alpha} \,, \tag{18}$$

where we have rewritten the last sum in terms of the generalized harmonic number $H_{n,m}$. The upper limit of the sum, $|X|$, is the maximum frequency rank, which coincides with the number of nonzero elements in the matrix X. This number is also equal to the maximum value of r in Eqn. (17) such that $X_{ij} \geq 1$, i.e., $|X| = k^{1/\alpha}$. Therefore we can write Eqn. (18) as,

$$|C| \sim |X|^\alpha \, H_{|X|,\alpha} \,. \tag{19}$$

We are interested in how $|X|$ is related to $|C|$ when both numbers are large; therefore we are free to expand the right hand side of the equation for large $|X|$. For this purpose we use the expansion of generalized harmonic numbers (Apostol, 1976),

$$H_{x,s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-s}) \quad \text{if} \quad s > 0,\, s \neq 1 \,, \tag{20}$$

giving,

$$|C| \sim \frac{|X|}{1-\alpha} + \zeta(\alpha)\, |X|^\alpha + O(1) \,, \tag{21}$$

where $\zeta(s)$ is the Riemann zeta function. In the limit that X is large, only one of the two terms on the right hand side of Eqn. (21) will be relevant, and which term that is depends on whether $\alpha > 1$,

$$|X| = \begin{cases} O(|C|) & \text{if } \alpha < 1, \\ O(|C|^{1/\alpha}) & \text{if } \alpha > 1. \end{cases} \tag{22}$$

For the corpora studied in this article, we observe that X_{ij} is well-modeled by Eqn. (17) with $\alpha = 1.25$. In this case we have that $|X| = O(|C|^{0.8})$. Therefore we conclude that the complexity of the model is much better than the worst case $O(V^2)$, and in fact it does somewhat better than the on-line window-based methods which scale like $O(|C|)$.

4 Experiments

4.1 Evaluation methods

We conduct experiments on the word analogy task of Mikolov et al. (2013a), a variety of word similarity tasks, as described in (Luong et al., 2013), and on the CoNLL-2003 shared benchmark dataset for NER (Tjong Kim Sang and De Meulder, 2003).

Table 2: Results on the word analogy task, given as percent accuracy. Underlined scores are best within groups of similarly-sized models; bold scores are best overall. HPCA vectors are publicly available[2]; (i)vLBL results are from (Mnih et al., 2013); skip-gram (SG) and CBOW results are from (Mikolov et al., 2013a,b); we trained SG† and CBOW† using the `word2vec` tool[3]. See text for details and a description of the SVD models.

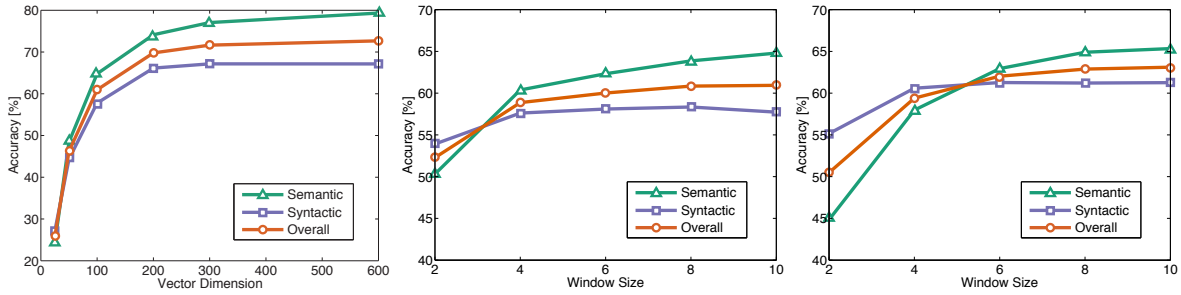
Model	Dim.	Size	Sem.	Syn.	Tot.
ivLBL	100	1.5B	55.9	50.1	53.2
HPCA	100	1.6B	4.2	16.4	10.8
GloVe	100	1.6B	<u>67.5</u>	<u>54.3</u>	<u>60.3</u>
SG	300	1B	61	61	61
CBOW	300	1.6B	16.1	52.6	36.1
vLBL	300	1.5B	54.2	<u>64.8</u>	60.0
ivLBL	300	1.5B	65.2	63.0	64.0
GloVe	300	1.6B	<u>80.8</u>	61.5	<u>70.3</u>
SVD	300	6B	6.3	8.1	7.3
SVD-S	300	6B	36.7	46.6	42.1
SVD-L	300	6B	56.6	63.0	60.1
CBOW†	300	6B	63.6	<u>67.4</u>	65.7
SG†	300	6B	73.0	66.0	69.1
GloVe	300	6B	<u>77.4</u>	67.0	<u>71.7</u>
CBOW	1000	6B	57.3	68.9	63.7
SG	1000	6B	66.1	65.1	65.6
SVD-L	300	42B	38.4	58.2	49.2
GloVe	300	42B	**<u>81.9</u>**	**<u>69.3</u>**	**<u>75.0</u>**

Word analogies. The word analogy task consists of questions like, "a is to b as c is to __?" The dataset contains 19,544 such questions, divided into a semantic subset and a syntactic subset. The semantic questions are typically analogies about people or places, like "Athens is to Greece as Berlin is to __?". The syntactic questions are typically analogies about verb tenses or forms of adjectives, for example "dance is to dancing as fly is to __?". To correctly answer the question, the model should uniquely identify the missing term, with only an exact correspondence counted as a correct match. We answer the question "a is to b as c is to __?" by finding the word d whose representation w_d is closest to $w_b - w_a + w_c$ according to the cosine similarity.[4]

[2] http://lebret.ch/words/

[3] http://code.google.com/p/word2vec/

[4] Levy et al. (2014) introduce a multiplicative analogy evaluation, 3CosMul, and report an accuracy of 68.24% on



(a) Symmetric context (b) Symmetric context (c) Asymmetric context

Figure 2: Accuracy on the analogy task as function of vector size and window size/type. All models are trained on the 6 billion token corpus. In (a), the window size is 10. In (b) and (c), the vector size is 100.

Word similarity. While the analogy task is our primary focus since it tests for interesting vector space substructures, we also evaluate our model on a variety of word similarity tasks in Table 3. These include WordSim-353 (Finkelstein et al., 2001), MC (Miller and Charles, 1991), RG (Rubenstein and Goodenough, 1965), SCWS (Huang et al., 2012), and RW (Luong et al., 2013).

Named entity recognition. The CoNLL-2003 English benchmark dataset for NER is a collection of documents from Reuters newswire articles, annotated with four entity types: person, location, organization, and miscellaneous. We train models on CoNLL-03 training data on test on three datasets: 1) ConLL-03 testing data, 2) ACE Phase 2 (2001-02) and ACE-2003 data, and 3) MUC7 Formal Run test set. We adopt the BIO2 annotation standard, as well as all the preprocessing steps described in (Wang and Manning, 2013). We use a comprehensive set of discrete features that comes with the standard distribution of the Stanford NER model (Finkel et al., 2005). A total of 437,905 discrete features were generated for the CoNLL-2003 training dataset. In addition, 50-dimensional vectors for each word of a five-word context are added and used as continuous features. With these features as input, we trained a conditional random field (CRF) with exactly the same setup as the CRF_{join} model of (Wang and Manning, 2013).

4.2 Corpora and training details

We trained our model on five corpora of varying sizes: a 2010 Wikipedia dump with 1 billion tokens; a 2014 Wikipedia dump with 1.6 billion tokens; Gigaword 5 which has 4.3 billion tokens; the combination Gigaword5 + Wikipedia2014, which has 6 billion tokens; and on 42 billion tokens of web data, from Common Crawl[5]. We tokenize and lowercase each corpus with the Stanford tokenizer, build a vocabulary of the 400,000 most frequent words[6], and then construct a matrix of co-occurrence counts X. In constructing X, we must choose how large the context window should be and whether to distinguish left context from right context. We explore the effect of these choices below. In all cases we use a decreasing weighting function, so that word pairs that are d words apart contribute $1/d$ to the total count. This is one way to account for the fact that very distant word pairs are expected to contain less relevant information about the words' relationship to one another.

For all our experiments, we set $x_{\max} = 100$, $\alpha = 3/4$, and train the model using AdaGrad (Duchi et al., 2011), stochastically sampling nonzero elements from X, with initial learning rate of 0.05. We run 50 iterations for vectors smaller than 300 dimensions, and 100 iterations otherwise (see Section 4.6 for more details about the convergence rate). Unless otherwise noted, we use a context of ten words to the left and ten words to the right.

The model generates two sets of word vectors, W and $\tilde{W}$. When X is symmetric, W and $\tilde{W}$ are equivalent and differ only as a result of their random initializations; the two sets of vectors should perform equivalently. On the other hand, there is evidence that for certain types of neural networks, training multiple instances of the network and then combining the results can help reduce overfitting and noise and generally improve results (Ciresan et al., 2012). With this in mind, we choose to use

the analogy task. This number is evaluated on a subset of the dataset so it is not included in Table 2. 3CosMul performed worse than cosine similarity in almost all of our experiments.

[5]To demonstrate the scalability of the model, we also trained it on a much larger sixth corpus, containing 840 billion tokens of web data, but in this case we did not lowercase the vocabulary, so the results are not directly comparable.

[6]For the model trained on Common Crawl data, we use a larger vocabulary of about 2 million words.

the sum $W + \tilde{W}$ as our word vectors. Doing so typically gives a small boost in performance, with the biggest increase in the semantic analogy task.

We compare with the published results of a variety of state-of-the-art models, as well as with our own results produced using the `word2vec` tool and with several baselines using SVDs. With `word2vec`, we train the skip-gram (SG†) and continuous bag-of-words (CBOW†) models on the 6 billion token corpus (Wikipedia 2014 + Gigaword 5) with a vocabulary of the top 400,000 most frequent words and a context window size of 10. We used 10 negative samples, which we show in Section 4.6 to be a good choice for this corpus.

For the SVD baselines, we generate a truncated matrix X_{trunc} which retains the information of how frequently each word occurs with only the top 10,000 most frequent words. This step is typical of many matrix-factorization-based methods as the extra columns can contribute a disproportionate number of zero entries and the methods are otherwise computationally expensive.

The singular vectors of this matrix constitute the baseline "SVD". We also evaluate two related baselines: "SVD-S" in which we take the SVD of $\sqrt{X_{\text{trunc}}}$, and "SVD-L" in which we take the SVD of $\log(1+X_{\text{trunc}})$. Both methods help compress the otherwise large range of values in X.[7]

4.3 Results

We present results on the word analogy task in Table 2. The GloVe model performs significantly better than the other baselines, often with smaller vector sizes and smaller corpora. Our results using the `word2vec` tool are somewhat better than most of the previously published results. This is due to a number of factors, including our choice to use negative sampling (which typically works better than the hierarchical softmax), the number of negative samples, and the choice of the corpus.

We demonstrate that the model can easily be trained on a large 42 billion token corpus, with a substantial corresponding performance boost. We note that increasing the corpus size does not guarantee improved results for other models, as can be seen by the decreased performance of the SVD-L model on this larger corpus. The fact that this basic SVD model does not scale well to large corpora lends further evidence to the necessity of the type of weighting scheme proposed in our model.

[7] We also investigated several other weighting schemes for transforming X; what we report here performed best. Many weighting schemes like PPMI destroy the sparsity of X and therefore cannot feasibly be used with large vocabularies. With smaller vocabularies, these information-theoretic transformations do indeed work well on word similarity measures, but they perform very poorly on the word analogy task.

Table 3: Spearman rank correlation on word similarity tasks. All vectors are 300-dimensional. The CBOW* vectors are from the `word2vec` website and differ in that they contain phrase vectors.

Model	Size	WS353	MC	RG	SCWS	RW
SVD	6B	35.3	35.1	42.5	38.3	25.6
SVD-S	6B	56.5	71.5	71.0	53.6	34.7
SVD-L	6B	65.7	72.7	75.1	56.5	37.0
CBOW†	6B	57.2	65.6	68.2	57.0	32.5
SG†	6B	62.8	65.2	69.7	58.1	37.2
GloVe	6B	65.8	72.7	77.8	53.9	38.1
SVD-L	42B	74.0	76.4	74.1	58.3	39.9
GloVe	42B	**75.9**	**83.6**	**82.9**	**59.6**	**47.8**
CBOW*	100B	68.4	79.6	75.4	59.4	45.5

Table 3 shows results on five different word similarity datasets. A similarity score is obtained from the word vectors by first normalizing each feature across the vocabulary and then calculating the cosine similarity. We compute Spearman's rank correlation coefficient between this score and the human judgments. CBOW* denotes the vectors available on the `word2vec` website that are trained with word and phrase vectors on 100B words of news data. GloVe outperforms it while using a corpus less than half the size.

Table 4 shows results on the NER task with the CRF-based model. The L-BFGS training terminates when no improvement has been achieved on the dev set for 25 iterations. Otherwise all configurations are identical to those used by Wang and Manning (2013). The model labeled *Discrete* is the baseline using a comprehensive set of discrete features that comes with the standard distribution of the Stanford NER model, but with no word vector features. In addition to the HPCA and SVD models discussed previously, we also compare to the models of Huang et al. (2012) (HSMN) and Collobert and Weston (2008) (CW). We trained the CBOW model using the `word2vec` tool[8]. The GloVe model outperforms all other methods on all evaluation metrics, except for the CoNLL test set, on which the HPCA method does slightly better. We conclude that the GloVe vectors are useful in downstream NLP tasks, as was first

[8] We use the same parameters as above, except in this case we found 5 negative samples to work slightly better than 10.

Table 4: F1 score on NER task with 50d vectors. *Discrete* is the baseline without word vectors. We use publicly-available vectors for HPCA, HSMN, and CW. See text for details.

Model	Dev	Test	ACE	MUC7
Discrete	91.0	85.4	77.4	73.4
SVD	90.8	85.7	77.3	73.7
SVD-S	91.0	85.5	77.6	74.3
SVD-L	90.5	84.8	73.6	71.5
HPCA	92.6	**88.7**	81.7	80.7
HSMN	90.5	85.7	78.7	74.7
CW	92.2	87.4	81.7	80.2
CBOW	93.1	88.2	82.2	81.1
GloVe	**93.2**	88.3	**82.9**	**82.2**

shown for neural vectors in (Turian et al., 2010).

4.4 Model Analysis: Vector Length and Context Size

In Fig. 2, we show the results of experiments that vary vector length and context window. A context window that extends to the left and right of a target word will be called symmetric, and one which extends only to the left will be called asymmetric. In (a), we observe diminishing returns for vectors larger than about 200 dimensions. In (b) and (c), we examine the effect of varying the window size for symmetric and asymmetric context windows. Performance is better on the syntactic subtask for small and asymmetric context windows, which aligns with the intuition that syntactic information is mostly drawn from the immediate context and can depend strongly on word order. Semantic information, on the other hand, is more frequently non-local, and more of it is captured with larger window sizes.

4.5 Model Analysis: Corpus Size

In Fig. 3, we show performance on the word analogy task for 300-dimensional vectors trained on different corpora. On the syntactic subtask, there is a monotonic increase in performance as the corpus size increases. This is to be expected since larger corpora typically produce better statistics. Interestingly, the same trend is not true for the semantic subtask, where the models trained on the smaller Wikipedia corpora do better than those trained on the larger Gigaword corpus. This is likely due to the large number of city- and country-based analogies in the analogy dataset and the fact that Wikipedia has fairly comprehensive articles for most such locations. Moreover, Wikipedia's

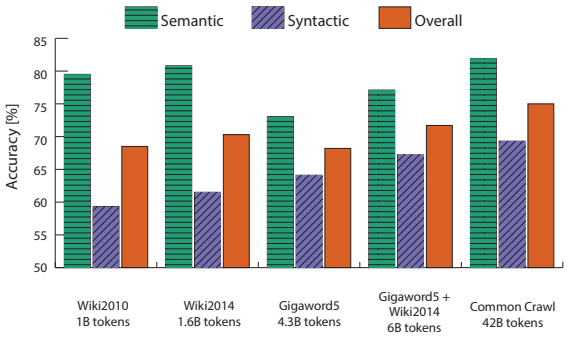


Figure 3: Accuracy on the analogy task for 300-dimensional vectors trained on different corpora.

entries are updated to assimilate new knowledge, whereas Gigaword is a fixed news repository with outdated and possibly incorrect information.

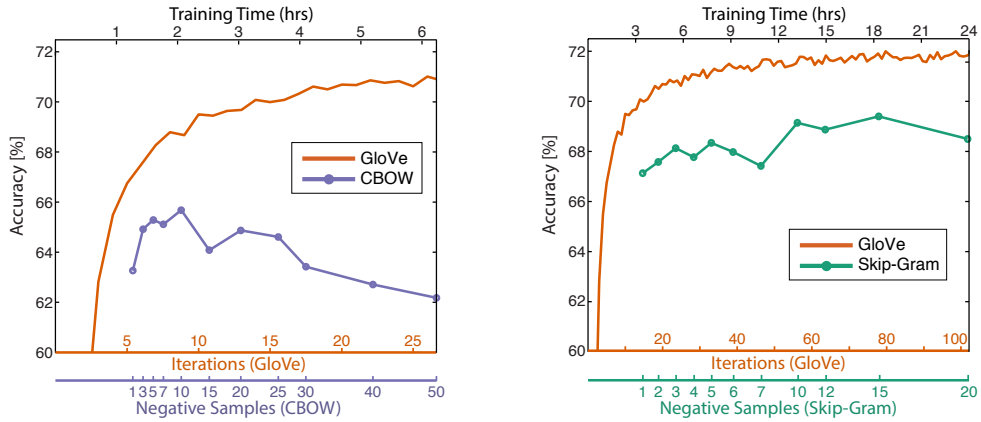
4.6 Model Analysis: Run-time

The total run-time is split between populating X and training the model. The former depends on many factors, including window size, vocabulary size, and corpus size. Though we did not do so, this step could easily be parallelized across multiple machines (see, e.g., Lebret and Collobert (2014) for some benchmarks). Using a single thread of a dual 2.1GHz Intel Xeon E5-2658 machine, populating X with a 10 word symmetric context window, a 400,000 word vocabulary, and a 6 billion token corpus takes about 85 minutes. Given X, the time it takes to train the model depends on the vector size and the number of iterations. For 300-dimensional vectors with the above settings (and using all 32 cores of the above machine), a single iteration takes 14 minutes. See Fig. 4 for a plot of the learning curve.

4.7 Model Analysis: Comparison with `word2vec`

A rigorous quantitative comparison of GloVe with `word2vec` is complicated by the existence of many parameters that have a strong effect on performance. We control for the main sources of variation that we identified in Sections 4.4 and 4.5 by setting the vector length, context window size, corpus, and vocabulary size to the configuration mentioned in the previous subsection.

The most important remaining variable to control for is training time. For GloVe, the relevant parameter is the number of training iterations. For `word2vec`, the obvious choice would be the number of training epochs. Unfortunately, the code is currently designed for only a single epoch:



(a) GloVe vs CBOW

(b) GloVe vs Skip-Gram

Figure 4: Overall accuracy on the word analogy task as a function of training time, which is governed by the number of iterations for GloVe and by the number of negative samples for CBOW (a) and skip-gram (b). In all cases, we train 300-dimensional vectors on the same 6B token corpus (Wikipedia 2014 + Gigaword 5) with the same 400,000 word vocabulary, and use a symmetric context window of size 10.

it specifies a learning schedule specific to a single pass through the data, making a modification for multiple passes a non-trivial task. Another choice is to vary the number of negative samples. Adding negative samples effectively increases the number of training words seen by the model, so in some ways it is analogous to extra epochs.

We set any unspecified parameters to their default values, assuming that they are close to optimal, though we acknowledge that this simplification should be relaxed in a more thorough analysis.

In Fig. 4, we plot the overall performance on the analogy task as a function of training time. The two x-axes at the bottom indicate the corresponding number of training iterations for GloVe and negative samples for `word2vec`. We note that `word2vec`'s performance actually decreases if the number of negative samples increases beyond about 10. Presumably this is because the negative sampling method does not approximate the target probability distribution well.[9]

For the same corpus, vocabulary, window size, and training time, GloVe consistently outperforms `word2vec`. It achieves better results faster, and also obtains the best results irrespective of speed.

5 Conclusion

Recently, considerable attention has been focused on the question of whether distributional word representations are best learned from count-based methods or from prediction-based methods. Currently, prediction-based models garner substantial support; for example, Baroni et al. (2014) argue that these models perform better across a range of tasks. In this work we argue that the two classes of methods are not dramatically different at a fundamental level since they both probe the underlying co-occurrence statistics of the corpus, but the efficiency with which the count-based methods capture global statistics can be advantageous. We construct a model that utilizes this main benefit of count data while simultaneously capturing the meaningful linear substructures prevalent in recent log-bilinear prediction-based methods like `word2vec`. The result, GloVe, is a new global log-bilinear regression model for the unsupervised learning of word representations that outperforms other models on word analogy, word similarity, and named entity recognition tasks.

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[9] In contrast, noise-contrastive estimation is an approximation which improves with more negative samples. In Table 1 of (Mnih et al., 2013), accuracy on the analogy task is a non-decreasing function of the number of negative samples.

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